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# LECTURE 6: Lennard-Jones Fluid Pair Potential
# Script: LJ
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# Description:
#   This notebook lets you interactively adjust epsilon and
↪   sigma of the LJ
#   potential and see how it changes the shape of the potential.
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# On Google Colab, install ipynb and enable interactive widgets
import sys
if "google.colab" in sys.modules:
    %pip install -q ipynb
    from google.colab import output
    output.enable_custom_widget_manager()

import numpy as np
import matplotlib.pyplot as plt
from ipywidgets import interact, FloatSlider
import ipywidgets as widgets

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Running locally.

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def plot_lennard_jones(epsilon_k, sigma):
    """
    Plots the Lennard-Jones potential.
    epsilon_k: Depth of the potential well in Kelvin (epsilon /
↪   kB)
    sigma: Distance where potential is zero (nm)
    """

    # 1. Define the range of r (distance)
    # We choose a range that covers most small molecules (0.25nm
↪   to 1.0nm)
    r = np.linspace(0.24, 1.0, 500)

    # 2. Calculate Potential Energy (in Kelvin)
    #  $V(r) = 4 * \epsilon_k * [ (\sigma/r)^{12} - (\sigma/r)^6 ]$ 
    # We plot V/kB, so we use epsilon_k directly
    V_over_kB = 4 * epsilon_k * ((sigma / r)**12 - (sigma /
↪   r)**6)

    # 3. Create Plot
    plt.figure(figsize=(8, 6))
    plt.plot(r, V_over_kB, linewidth=3, color='blue', label='LJ
↪   Potential')

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# 4. Mark the Minimum (Equilibrium separation)
# The minimum occurs at  $r = 2^{(1/6)} * \sigma$ 
r_min = (2**(1/6)) * sigma
V_min = -epsilon_k

plt.plot(r_min, V_min, 'ro', markersize=10,
         label=f'Well Depth ( $\epsilon/k_B$ ) = -{epsilon_k}
↪ K')

# 5. Formatting
plt.axhline(0, color='black', linestyle='--', linewidth=1)
plt.ylim(-400, 200) # Fixed Y-limit to allow visual
↪ comparison of well depths
plt.xlim(0.24, 1.0)
plt.xlabel('Intermolecular Distance  $r$  (nm)', fontsize=12)
plt.ylabel('Potential Energy /  $k_B$  (K)', fontsize=12)
plt.title(f'Lennard-Jones Potential:  $\epsilon/k_B$  =
↪ {epsilon_k:.0f} K,  $\sigma$  = {sigma:.3f} nm', fontsize=14)
plt.grid(True, which='both', linestyle='--', alpha=0.5)
plt.legend(fontsize=11)

# N. B.  $T_{\text{boiling}}$  is roughly  $0.7 * \epsilon/k_B$  for noble
↪ gases

plt.show()

# Create the sliders
# Ranges cover Hydrogen (small) to Xenon (large)
interact(plot_lennard_jones,
        epsilon_k=FloatSlider(value=120, min=30, max=300,
↪ step=10,
                                description='epsilon/kB (K)'),
        sigma=FloatSlider(value=0.34, min=0.25, max=0.45,
↪ step=0.01,
                                description='sigma (nm)'));

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interactive(children=(FloatSlider(value=120.0, description='epsilon/kB (K)

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# Static preview: results for the default inputs above.
# The output is saved so the course website can show it.
# To try other inputs, use the controls above, or open
# this notebook in Colab (badge on the course website).
plot_lennard_jones(epsilon_k=120, sigma=0.34)

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